

MTH-356: Linear Algebra with Application

Lecture 2 – Vectors and Their Role in Machine Learning

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📅 Week 1

Topic: Vectors and Their Role in Machine Learning

Duration: 1.5 hours lecture + 1 hour tutorial

🔑 Why This Matters for Math Students

Vectors are the language of data. In machine learning, every sample, image, sound, or text input can be represented as a vector of numerical values. Understanding vectors and their operations allows us to see how data is transformed, compared, and modeled mathematically.

1 Concept Overview

A **vector** is an ordered collection of numbers representing either a point or a direction in space. Mathematically,

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n$$

Each element x_i is called a *component* of the vector.

For example:

$$\mathbf{x} = \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix} \in \mathbb{R}^3$$

represents a point in 3D space.

🏠 Machine Learning Applications

In ML:

- Each data sample is a vector of features.
- Each word in NLP is represented as a vector embedding.
- Model parameters (weights) are stored as vectors.

2 Vectors in Machine Learning Context

2.1 Feature Representation

A data point with n features is represented as a vector:

$$\mathbf{x} = [x_1, x_2, \dots, x_n]^T$$

Example: Predicting house price

$$\mathbf{x} = \begin{bmatrix} 120 \\ 3 \\ 200 \end{bmatrix} \Rightarrow \text{Size (m}^2\text{), Bedrooms, Price (in \$1000s)}$$

When we have multiple samples:

$$X = \begin{bmatrix} 120 & 3 & 200 \\ 150 & 4 & 250 \\ 90 & 2 & 150 \end{bmatrix}$$

Each row $\rightarrow$ one sample, Each column $\rightarrow$ one feature.

2.2 Feature Weights and Predictions

A simple linear model:

$$\hat{y} = \mathbf{w}^T \mathbf{x} + b$$

where

- $\mathbf{w}$ = weight vector
- $\mathbf{x}$ = feature vector
- b = bias term

Interpretation: The dot product $\mathbf{w}^T \mathbf{x}$ measures the influence of each feature on the output.

3 Vector Operations and Geometry

3.1 Vector Addition

$$\mathbf{a} + \mathbf{b} = \begin{bmatrix} a_1 + b_1 \\ a_2 + b_2 \\ \vdots \\ a_n + b_n \end{bmatrix}$$

Example:

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} + \begin{bmatrix} -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$$

ML Relevance: Combining signals, gradients, or feature vectors.

3.2 Scalar Multiplication

$$c\mathbf{a} = \begin{bmatrix} ca_1 \\ ca_2 \\ \vdots \\ ca_n \end{bmatrix}$$

Example: $2 \times [3, -2]^T = [6, -4]^T$

ML Relevance: Feature scaling, normalization.

3.3 Magnitude (Length)

$$\|\mathbf{a}\| = \sqrt{a_1^2 + a_2^2 + \cdots + a_n^2}$$

Example:

$$\left\| \begin{bmatrix} 3 \\ 4 \end{bmatrix} \right\| = 5$$

ML Relevance: Used in normalization and distance metrics.

3.4 Dot Product

$$\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + \cdots + a_nb_n = \|\mathbf{a}\| \|\mathbf{b}\| \cos \theta$$

Geometric Meaning: Measures similarity between two vectors.

Machine Learning Applications

Cosine Similarity:

$$\text{similarity}(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|}$$

Used in:

- Text similarity (NLP)
- Recommendation systems
- Word embedding comparisons

4 Worked ML Example – Cosine Similarity in Text

Let

$$\mathbf{d}_1 = [1, 2, 0, 3], \quad \mathbf{d}_2 = [0, 1, 1, 2]$$

Compute:

$$\mathbf{d}_1 \cdot \mathbf{d}_2 = 8, \quad \|\mathbf{d}_1\| = \sqrt{14}, \quad \|\mathbf{d}_2\| = \sqrt{6}$$

$$\text{Cosine Similarity} = \frac{8}{\sqrt{14}\sqrt{6}} \approx 0.87$$

⇒ Documents are highly similar ($0.87 \approx 1$).

Theoretical Foundations

Key Geometric Ideas:

- Vectors represent directions and magnitudes.
- Orthogonal vectors ($\mathbf{a} \cdot \mathbf{b} = 0$) are independent.
- The span of a set of vectors forms a subspace.

These ideas later connect to PCA, regression, and neural networks.

Career Pathways & Future Scope

Skills Built:

- Understanding data as vectors in high-dimensional space.
- Computing similarity and projections.
- Interpreting model weights geometrically.

Next Lecture: Matrix Multiplication – Neural Network Layers as Linear Transformations.

Learning Resources

Textbook:

- Deisenroth, M. P., Faisal, A. A., & Ong, C. S. (2020). *Mathematics for Machine Learning*. Cambridge University Press.

Practice: Use Python (NumPy) to compute vector operations and visualize them using `matplotlib`.